2. Dynamics of Linear Systems

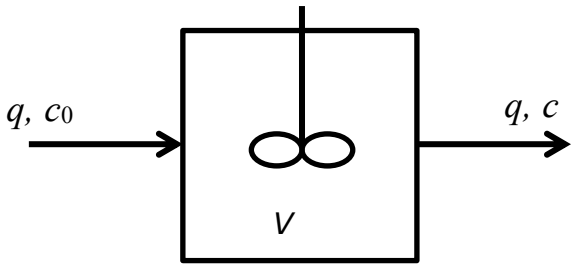
2.1 Introduction

In order to see the relationships between input and output variables, allowing the selection of appropriate manipulated variables, an understanding of the system dynamics is needed. It is surprising how often poor control is designed in industry because people do not take the time to understand the process. The starting point is writing the appropriate dynamic heat and material balances, as introduced in CEBT IB Heat and Mass Transfer and Engineering Mathematics; this section starts with a review**.**

The advent of more powerful desktop computing means that dynamic process models can now be easily built using commercial software packages.

2.2 Dynamic Balances

2.2.1 CST



A steady volumetric flow rate q is fed to a well-stirred tank of constant volume V. The concentration of an inert (and dilute) tracer in the feed is c_0, and in the tank is c. The material balance on the tracer is

$$qc_0 = V\frac{dc}{dt} + qc$$

which can be written as

$$\tau\frac{dc}{dt} + c = c_0$$

where $\tau = V/q$ is the residence time. This equation is sometimes known as a first order lag, and crops up a lot.

As an example take

$$t < 0: c = c_0 = 0$$
$$t \geq 0: c_0 = c_{0s} = \text{constant}$$

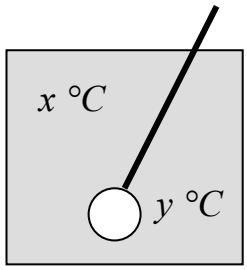
This can be solved by a variety of methods (Exercise – find several), to give

$$c = c_{0s}\left(1 - e^{-t/\tau}\right)$$

Note that $c \rightarrow c_{0s}$ as $t \rightarrow \infty$, with time constant τ. While the response starts to change as soon as the input changes, it takes several periods of τ to approach the new value c_{0s} – the response lags behind the input.

Example 2.1 Thermocouple in a liquid-filled bath

Another example of a first order lag is the thermocouple. For example, the hot junction of a thermocouple, or the bulb of a thermometer, is immersed in a large, well-stirred liquid-filled bath at a uniform, and constant (approximately, because it's large) temperature *x* °C. The reading on the thermocouple is *y* °C. The mass of the thermocouple is *m*, its heat capacity is c_p, the heat transfer coefficient is *h* and the area for heat transfer is *A*.



$$mc_p \frac{dy}{dt} = hA(x - y)$$

2.2.2 CSTR

This is the same as 2.2.1, but here the dilute component decays via a first order reaction with rate constant *k*. The material balance is

$$qc_0 = V\frac{dc}{dt} + qc + kVc \rightarrow \tau\frac{dc}{dt} + (1 + k\tau)c = c_0$$

or

$$\frac{\tau}{1 + k\tau}\frac{dc}{dt} + c = \frac{c_0}{1 + k\tau}$$

another first order lag, with time constant $\tau/(1 + k\tau)$. For the case

$$t < 0: c = c_0 = 0$$
$$t \geq 0: c_0 = c_{0s} = \text{constant}$$

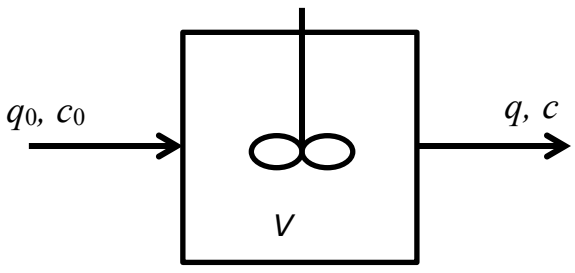
The solution is

$$c = \frac{c_{0s}}{1 + k\tau}\left(1 - e^{-(1+k\tau)t/\tau}\right)$$

As $t \rightarrow \infty$, $c \rightarrow c_{0s}/(1 + k\tau)$ – see CEBT IB Reaction Engineering.

2.2.3 CSTR with varying volume

This is the same as 2.2.2, but with inlet and outlet flow rates different.



The volume balance is

$$\frac{dV}{dt} = q_0 - q$$

and the material balance is

$$q_0 c_0 = \frac{d}{dt}(Vc) + qc + kVc = \frac{dV}{dt}c + V\frac{dc}{dt} + qc + kVc$$

Using the volume balance gives

$$V\frac{dc}{dt} + (q_0 + kV)c = q_0 c_0$$

The integration of this depends on the how q_0 and q depend on time. First find V:

$$V(t) - V(0) = \int_0^t \left(q_0(t') - q(t')\right) dt'$$

and then find c.

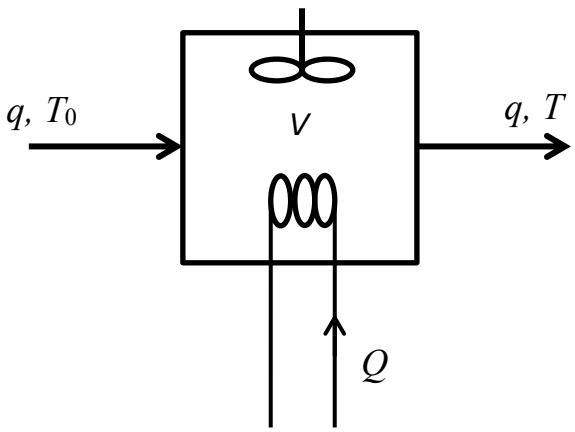
2.2.4 Impulse input

Consider 2.2.3, when at some time a pulse of volume V_p and concentration c_p is added to the tank. The simplest way to proceed is to apply volume and material balances over the pulse. If the tank has volume V_1 and concentration c_1 just before the pulse, and volume V_2 and concentration c_2 just after the pulse, then

$$V_1 + V_p = V_2, \quad V_1 c_1 + V_p c_p = V_2 c_2$$

Differential balances can be used away from the pulse.

2.2.5 Unsteady flow energy equation - liquids



Q is the rate of heating, and q is volumetric flow rate. For this example take the volumetric flow rate, volume, and specific heat capacity to be constant.

The energy balance is

$$\frac{dU}{dt} = \rho q c_p T_0 - \rho q c_p T + Q$$

For liquids,

$$\frac{dU}{dt} \approx \frac{dH}{dt} = \rho V c_p \frac{dT}{dt}$$

so that

$$\rho V c_p \frac{dT}{dt} = \rho q c_p T_0 - \rho q c_p T + Q \rightarrow \tau \frac{dT}{dt} + T = T_0 + \frac{Q}{\rho q c_p}$$

where $\tau = V/q$.

For example take

$$t < 0: Q = 0, T = T_0 = T_{0s} = \text{const}$$
$$t \geq 0: Q = Q_s = \text{const}, T_0 = T_{0s} = \text{const}$$

This is a first order lag with time constant τ. When $t = 0$, $T = T_{0s}$, and when $t \rightarrow \infty$, $T \rightarrow T_{0s} + Q_s/\rho q c_p$, so

$$T = T_{0s} + \frac{Q_s}{\rho q c_p} - \frac{Q_s}{\rho q c_p} e^{-t/\tau}$$

The first two terms on the right-hand-side give the asymptotic value, and the third term fixes up the initial condition.

2.2.6 Unsteady state energy equation – gases

An ideal gas discharges from a well-mixed insulated vessel of constant volume V at mass flow rate w. The mass of gas in the vessel is m, the pressure is p, and the temperature is T; the initial values are denoted by subscript 0. Assume that the gas leaving the tank has the same specific enthalpy as the gas in the tank. The mass balance is

$$\frac{dm}{dt} = -w$$

and the unsteady flow energy equation is, using $\gamma = c_p/c_V$,

$$\frac{dU}{dt} = \frac{d}{dt}(c_V m T) = -w c_p T \rightarrow \frac{d}{dt}(mT) = -\gamma w T \rightarrow \frac{d}{dm}(mT) = \gamma T$$

$$\rightarrow T + m\frac{dT}{dm} = \gamma T \rightarrow m\frac{dT}{dm} = (\gamma - 1)T$$

$$\rightarrow \frac{dT}{T} = (\gamma - 1)\frac{dm}{m} \rightarrow [\ln T]_{T_0}^{T} = (\gamma - 1)[\ln m]_{m_0}^{m} \rightarrow T = T_0 \left(\frac{m}{m_0}\right)^{\gamma - 1}$$

Now

$$pV = mRT \rightarrow \frac{m}{m_0} = \frac{p}{p_0}\frac{T_0}{T} \rightarrow \frac{T}{T_0} = \left(\frac{p}{p_0}\right)^{(\gamma-1)/\gamma} = \left(\frac{m}{m_0}\right)^{\gamma - 1}$$

What does this imply?

Note that we have taken the reference temperature for *U* to be 0 K. What is then the reference temperature for *H*? What would be the consequence of taking a different reference temperature for *U*?

2.3 Linear Models and Transfer Functions

Simple process systems can often be represented by dynamic heat and material balances, written as a set of simultaneous linear algebraic and differential equations that can be solved analytically using Laplace transforms:

$$LT[y(t)] = \bar{y}(s) = \int_0^\infty y(t)e^{-st}dt$$

There is a table in the Department Data Book. The resulting equations are algebraic and hence easy to manipulate.

Process control is concerned with how the output variables from a system are affected by a change in the input variables. A convenient way of modelling this behaviour is through the use of *transfer functions* and *block diagrams*. As will be seen later, having the system represented in the *s*-domain of the Laplace transform allows us to gain a valuable insight into the system behaviour without determining the complete transient solution.

An important constraint is that the set of differential equations must be linear. If they are not, they can be linearised to give approximate results in the region of a steady state - as process control is generally interested in operation over a narrow range of process parameters, this is generally OK.

This section introduces the transfer function which is a fundamental building block for all process control problems as it contains all the steady-state and dynamic information about a process in a compact form which is convenient to manipulate.

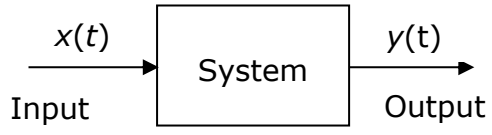
For process control purposes systems can be categorised as:

- Single Input- Single Output (SISO) – one input variable affects one output variable.
- Multiple Input-Multiple Output (MIMO) – multiple input variables will affect multiple output variables.

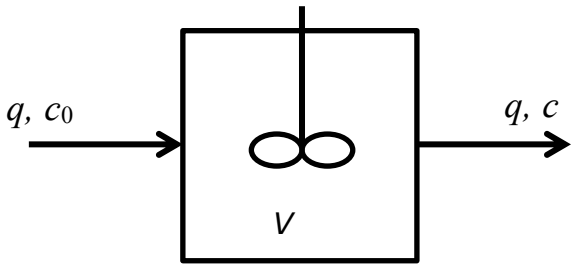
First, the transfer function and block diagram will be introduced using an example of a SISO.

2.4 Transfer Function for a Single Input – Single Output System (SISO)

A simple representation of a SISO is shown below where an output variable in the time domain has a dependency on an input variable in the time domain.



Example 2.2 Constant volume mixing tank



Initially $c = c_0 = 0$. Assume that the vessel is perfectly mixed, the density of the solute and solvent are equal and q is constant. We will formulate a model that relates the inlet and outlet concentration of the solute.

The material balance on the solute is (as in 2.2.1)

$$qc_0 = V\frac{dc}{dt} + qc \rightarrow \tau\frac{dc}{dt} + c = c_0, \text{ where } \tau = \frac{V}{q}$$

Take Laplace transforms (see Department Data Book):

$$\tau[s\bar{c} - c(0)] + \bar{c} = \bar{c}_0$$

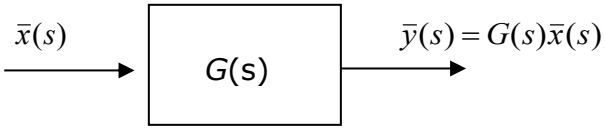
Here $c(0) = 0$, so

$$\frac{\bar{c}}{\bar{c}_0} = \frac{1}{\tau s + 1} = G(s) \text{ the transfer function}$$

The transfer function of a system is defined as the Laplace transform of the output variable $y(t)$ divided by the Laplace transform of the input variable $x(t)$, with all initial conditions equal to zero:

$$\text{Transfer function} = G(s) = \frac{\bar{y}(s)}{\bar{x}(s)}$$

The block diagram is a simple way of representing transfer functions in the Laplace domain and is shown below:



The block diagram provides a method for combining individual transfer functions into an overall transfer function for more complex systems. We will return to this later.

2.5 Deviation Variables

The transfer function definition assumes initial conditions are equal to zero. To ensure that this criterion is met, dynamic behaviour is measured in terms of deviations from a steady-state.

In Example 2.2 suppose that instead the initial values are $c_0 = c = 5$ kmol/m^3. Then

$$\tau[s\bar{c} - 5] + \bar{c} = \bar{c}_0$$

and we cannot find $\bar{c}/\bar{c}_0$.

However, if we express the variables as *deviation variables*:

$$c'(t) = c(t) - 5, \;\; c_0'(t) = c_0(t) - 5$$

the ODE becomes

$$\tau\frac{d(5+c')}{dt} + (5+c') = (5+c_0') \rightarrow \tau\frac{dc'}{dt} + c' = c_0' \rightarrow \frac{\bar{c}}{\bar{c}_0} = \frac{1}{\tau s + 1} = G(s)$$

Here we have used the convention

$$\bar{c} = LT[c'], \;\; \bar{c}_0 = LT[c_0']$$

Note that the steady state cancels from the ODE – it must because the steady state, which has zero deviation variables, satisfies the ODE.

In process control we will only use deviation variables:

$$x' = x - x_s \;\text{ and }\; y' = y - y_s$$

where the subscript *s* denotes steady state. We will also use the convention

$$\bar{y}(s) = L[y'] = L[y - y_s] \;\; etc$$

2.6 Transfer Function Assumptions

There are three major assumptions employed in using transfer functions:

1. Time always begins at $t = 0$.
2. Zero initial conditions (steady state) are assumed for all systems, e.g.

$$x' = \frac{dx'}{dt} = 0 \;\text{ at }\; t = 0$$

3. A transfer function relates one output to one input.

2.7 Elementary Transfer Functions

2.7.1 First Order Systems

In general, material or energy balances on “lumped” systems are described by a first order ODE of the general form:

$$\tau\frac{dy}{dt} + y = K_p x$$

where τ = time constant, K_P = process gain.

Using deviation variables and taking the Laplace transform:

$$\tau s\bar{y} + \bar{y} = K_p\bar{x}$$

$$\frac{\bar{y}}{\bar{x}} = G(s) = \frac{K_p}{\tau s + 1}$$

The process gain is the steady state gain when the output variable reaches a steady state after an input disturbance which becomes constant as $t \to \infty$. It is a measure of how a permanent change in the input affects the output in the long term. We can see this by looking at the ODE or by using Laplace Transforms.

In terms of the ODE:

$$\tau\frac{dy'}{dt} + y' = K_p x'$$

Suppose that at $t = 0$, x' changes stepwise from zero to Δx. If y' eventually reaches a steady value, Δy, the derivative term in the ODE is zero, and so

$$\Delta y = K_p\Delta x$$

We can find this too using the Final Value Theorem of Laplace transforms, which is, if the limits exist,

$$\lim_{t\to\infty} y'(t) = \lim_{s\to 0}\ s\bar{y}(s)$$

So with

$$\bar{y} = G(s)\bar{x} \text{ and } \bar{x} = \frac{\Delta x}{s}$$

The FVT gives

$$\Delta y = \lim_{s\to 0} sG(s)\frac{\Delta x}{s} = \Delta x G(0)$$

so that

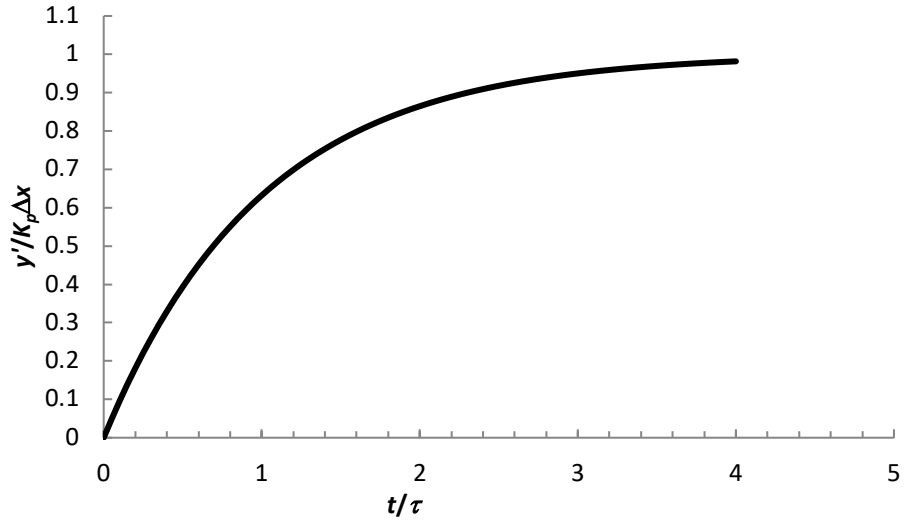
$$\frac{\Delta y}{\Delta x} = G(0) = K_p$$

The transfer function contains all the information necessary to determine how an output responds to a disturbance in the input. The output response to the step input of magnitude Δx is found from

$$\bar{y} = G(s)\bar{x} = \frac{K_p}{\tau s + 1}\frac{\Delta x}{s} = K_p\Delta x\left[\frac{1}{s} - \frac{\tau}{\tau s + 1}\right]$$

$$\to y' = K_p\Delta x\left[1 - e^{-t/\tau}\right]$$

The partial fractions can be found using the Cover-Up Rule. The so-called step response of the first order lag is plotted below.



1. The output starts to change immediately.
2. The characteristic time τ governs the speed of response.
3. The gradient at the origin is

$$\frac{dy'}{dt} = \frac{K_p \Delta x}{\tau} \text{ at } t = 0$$

4. The tangent at the origin cuts the asymptotic value at $t = \tau$.
5. The gain controls whether the system amplifies or attenuates the input disturbance.

As well as the Heaviside step input, the impulse, or Dirac delta-function is used as a test input. This is a pulse of infinite magnitude occurring over an infinitesimal time, with unit area. You can think of the delta function at the origin as the limit of Gaussian distributions centred at the origin as $\sigma \to 0$, or as the limit of rectangles of height n and width $1/n$ as $n \to \infty$.

$$LT[\delta(t)] = 1$$

So if $x(t) = \delta(t)$,

$$\bar{y} = G(s)\bar{x} = G(s)$$

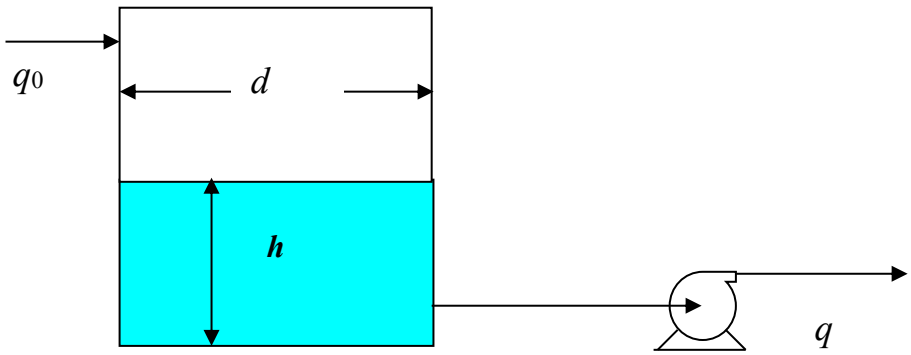
The Laplace transform of the impulse response is the transfer function. So for the first order lag

$$\bar{y} = G(s)\bar{x} = \frac{K_p}{\tau s + 1} \to y' = \frac{K_p}{\tau} e^{-t/\tau}$$

The delta function can be thought of as the derivative of the unit step function (think of the Fundamental Theorem of Calculus), so the response to the delta function input is the derivative of the response to the unit step function input – try it.

Integrator

A special type of first order system is the integrator, an example of which is a tank level where the flow into or out of the tank is **not** dependent on the level in the tank (see 2.2.3):



A volume balance on the tank gives

$$\frac{\pi d^2}{4}\frac{dh}{dt} = q_0 - q$$

If q_0 and q are independent of the level in the tank, for example if q_0 is discharge from an upstream process, and q is pumped out of the vessel, then h can be found by integration.

The general form of the integrator is

$$T_H \frac{dy}{dt} = x \rightarrow T_H \frac{dy'}{dt} = x' \rightarrow y' = \frac{1}{T_H}\int_0^t x'(t')dt'$$

where T_H is the holdup time. This system accumulates the net input, hence its name.

Take the Laplace transform:

$$T_H s\bar{y} = \bar{x} \rightarrow \frac{\bar{y}}{\bar{x}} = G(s) = \frac{1}{T_H s}$$

For a unit step input,

$$\bar{x} = \frac{1}{s} \rightarrow \bar{y} = \frac{1}{T_H s^2} \rightarrow y' = \frac{t}{T_H}$$

No new steady state is reached: the system is not self-regulating, it is unstable, the response is unbounded. Note: all non-self-regulating variables, e.g. levels, must be controlled.

For a delta function input,

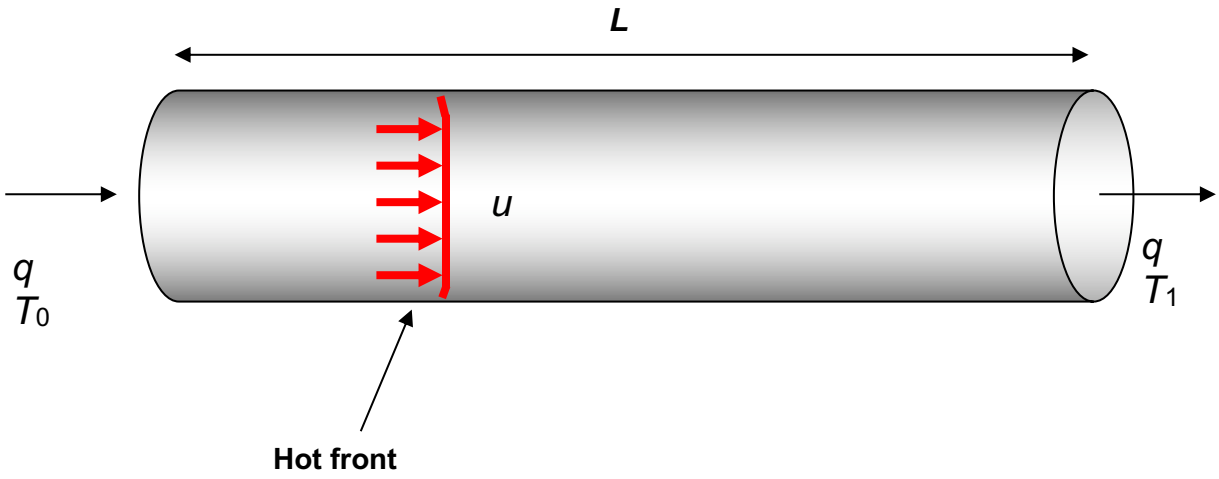
$$\bar{x} = 1 \rightarrow \bar{y} = \frac{1}{T_H s} \rightarrow y' = \frac{1}{T_H}$$

The integrator integrates the impulse, and after that *y'* stays constant.

2.7.2 Dead Time

Rarely do things occur instantaneously in process plants, for example items of equipment are connected by piping that results in a time delay, or dead time, between material entering one end of the pipe and coming out the other.

Example 2.3 Plug flow device



Assuming that the velocity, *u* is uniform over a cross section, and along the pipe, the time for fluid to pass through the vessel is $t_d = L/u$. Then

$$T_1(t) = T_0(t - t_d) \text{ i.e. } y'(t) = x'(t - t_d)$$

$$\bar{y} = \int_0^\infty y'(t)e^{-st}dt = \int_0^\infty x'(t - t_d)e^{-st}dt = \int_{t_d}^\infty x'(t - t_d)e^{-st}dt = \bar{x}e^{-t_d s} \rightarrow G(s) = e^{-t_d s}$$

where we have used the fact that $x'(t) = 0$ if $t < 0$.

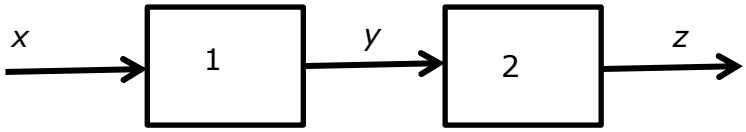
A step input gives a step output a time t_d later. A delta function input gives a delta function output a time t_d later.

2.8 Second-Order Linear Systems

Second order systems occur when two first order or one second order ordinary differential equation is needed to model the dynamic behaviour. An example from mechanics is the vertical motion of a point mass *m* suspended by an elastic string with elastic force *k* times extension, in a medium which provides a resistance to motion μ times velocity. The equation of motion is, where *y* is downward displacement from equilibrium position (Exercise):

$$m\ddot{y} = -\mu\dot{y} - ky$$

Another example is 2 CSTs in series:



$$\tau_1\dot{y} + y = x,\ \ \tau_2\dot{z} + z = y \ \ \rightarrow \ \ \tau_1\tau_2\ddot{z} + (\tau_1 + \tau_2)\dot{z} + z = x$$

The general second-order ODE with constant coefficients (i.e. a_2, a_1 , a_0 and b are constants) is:

$$a_2\frac{d^2y}{dt^2} + a_1\frac{dy}{dt} + a_0y = bx$$

or, rewriting in the standard form $\left(a_0 \neq 0\right)$

$$\tau^2\ddot{y} + 2\zeta\tau\dot{y} + y = K_px$$

where

$$\tau^2 = \frac{a_2}{a_0},\ \ 2\zeta\tau = \frac{a_1}{a_0},\ \ K_p = \frac{b}{a_0}$$

The constants for the system are:

τ = natural period of oscillation (a name – the system oscillates with this period only if ζ = 0)

ζ = damping factor

K_p = steady-state gain

Take deviation variables and Laplace transform:

$$G(s) = \frac{\bar{y}}{\bar{x}} = \frac{K_p}{\tau^2s^2 + 2\zeta\tau s + 1} = \frac{K_p/\tau^2}{(s - p_1)(s - p_2)}$$

where p_1 and p_2 are the poles of G; they are the solutions of the *characteristic equation* (the equation you get by putting the denominator of G equal to zero):

$$\tau^2s^2 + 2\zeta\tau s + 1 = 0 \rightarrow p_1, p_2 = \frac{-\zeta \pm \sqrt{\zeta^2 - 1}}{\tau}$$

The nature of the poles and, as we will see, the response, depend on ζ:

$\zeta > 1 \rightarrow$ p_1 and p_2 are real and negative

$|\zeta| < 1 \rightarrow$ p_1 and p_2 are complex

$\zeta < -1 \rightarrow$ p_1 and p_2 are real and positive

2.9 Unit step response

$$x'(t) = H(t) \rightarrow \bar{x} = \frac{1}{s}$$

If $p_1 \neq p_2$

$$\bar{y} = \frac{K_p/\tau^2}{s(s - p_1)(s - p_2)} = \frac{K_p}{\tau^2}\left[\frac{1}{p_1p_2s} + \frac{1}{p_1(p_1 - p_2)(s - p_1)} + \frac{1}{p_2(p_2 - p_1)(s - p_2)}\right]$$

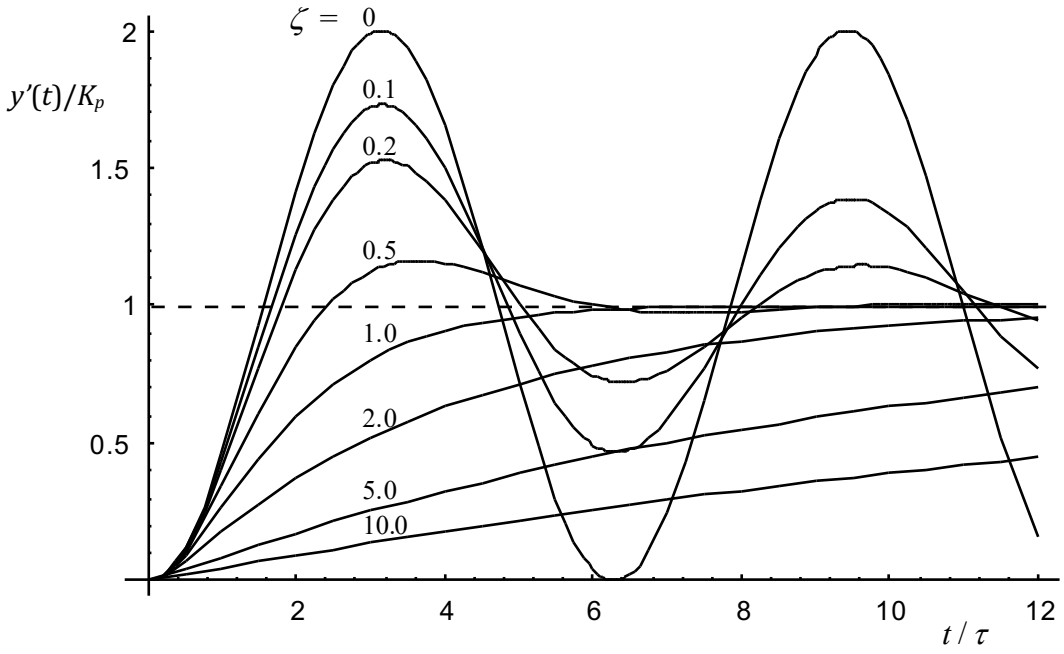
where the Cover-Up Rule has been used to find the partial fractions. The expression for $\bar{y}$ can be simplified using

$$p_1 p_2 = \frac{1}{\tau^2}$$

to give

$$\bar{y} = K_p\left[\frac{1}{s} - \frac{1}{p_1 - p_2}\left(\frac{p_1}{s - p_2} - \frac{p_2}{s - p_1}\right)\right] \;\rightarrow\; y' = K_p\left[1 - \frac{1}{p_1 - p_2}(p_1 e^{p_2 t} - p_2 e^{p_1 t})\right]$$

Now let's examine the unit step response for various values of the damping factor, ζ.



Overdamped systems: $\zeta > 1$

- p_1 and p_2 are real and negative
- stable response
- sluggish response
- as $t \rightarrow \infty$, $y \rightarrow K_p$

Critically damped systems: $\zeta = 1$

$$p_1, p_2 = -\frac{1}{\tau}$$

$$\bar{y} = \frac{K_p/\tau^2}{s\left(s + \frac{1}{\tau}\right)^2} = K_p\left[\frac{1}{s} - \frac{1}{s + \frac{1}{\tau}} - \frac{1}{\tau\left(s + \frac{1}{\tau}\right)^2}\right] \rightarrow y' = K_p\left[1 - \left(1 + \frac{t}{\tau}\right)e^{-t/\tau}\right]$$

- $p_1 = p_2$ real and negative
- stable response

- faster response than for $\zeta > 1$

Underdamped systems: $0 < \zeta < 1$

$$p_1, p_2 = \frac{-\zeta \pm i\sqrt{1-\zeta^2}}{\tau}$$

The poles p_1 and p_2 are complex, but there is stable response because $\zeta > 0$. Write

$$\omega = \frac{\sqrt{1-\zeta^2}}{\tau}$$

and then

$$e^{p_1 t} = e^{-\zeta t/\tau}(\cos(\omega t) + i\sin(\omega t)), \quad e^{p_2 t} = e^{-\zeta t/\tau}(\cos(\omega t) - i\sin(\omega t))$$

$$y' = K_p\left[1 - \frac{\tau}{2i\sqrt{1-\zeta^2}} e^{-\zeta t/\tau}\left\{\left(\frac{-\zeta + i\sqrt{1-\zeta^2}}{\tau}\right)(\cos(\omega t) - i\sin(\omega t)) - \left(\frac{-\zeta - i\sqrt{1-\zeta^2}}{\tau}\right)(\cos(\omega t) + i\sin(\omega t))\right\}\right]$$

The imaginary bits cancel out, as they must, leaving

$$y' = K_p\left[1 - e^{-\zeta t/\tau}\left(\cos(\omega t) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin(\omega t)\right)\right] = K_p\left[1 - \frac{e^{-\zeta t/\tau}}{\sqrt{1-\zeta^2}}\sin(\omega t + \phi)\right]$$

where

$$\phi = \tan^{-1}\left[\frac{\sqrt{1-\zeta^2}}{\zeta}\right]$$

- stable response
- complex poles lead to oscillations
- overshoot

Undamped systems: $\zeta = 0$

When the damping coefficient is set to zero the characteristic equation becomes:

$$\tau^2 s^2 + 1 = 0$$

- poles are imaginary
- there is no decay of the oscillating terms.
- the response oscillates at the frequency $1/\tau$ rad/s and period $T = 2\pi\tau$

Unstable systems: $\zeta < 0$

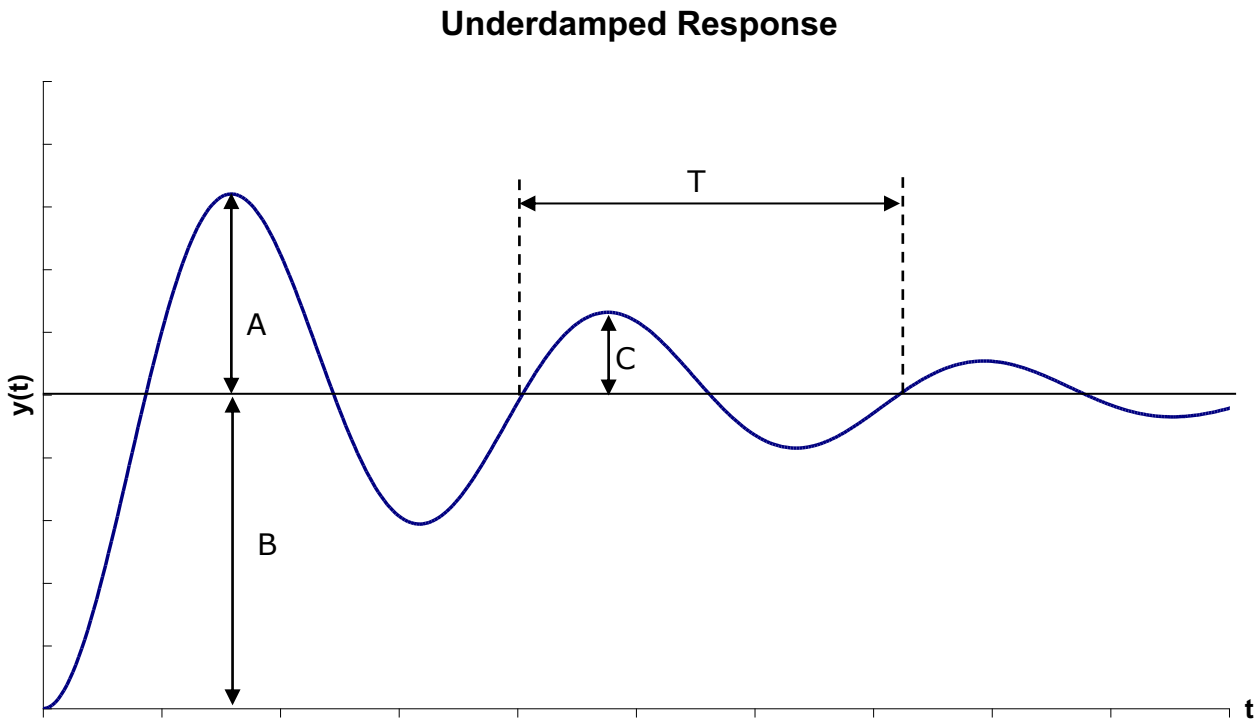
If the damping coefficient is negative, the exponential term increases without bound and hence the system is unstable.

- for-1< ζ<0, the roots are complex, with positive real part:

$$p_1 = -\frac{\zeta}{\tau} + i\frac{\sqrt{1-\zeta^2}}{\tau}, p_2 = -\frac{\zeta}{\tau} - i\frac{\sqrt{1-\zeta^2}}{\tau}$$

- for ζ< –1, the roots are real and positive.

2.10 Characteristics of the underdamped response to step input



Exercise: Use the results above to show that the following relationships hold.

Overshoot $$\frac{A}{B} = \exp\left[\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right]$$

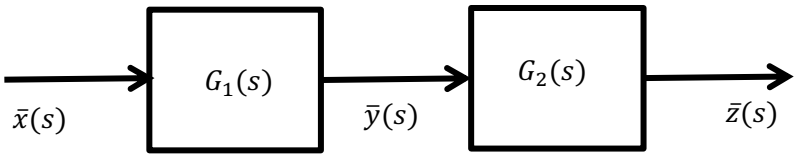
Decay ratio $$\frac{C}{A} = \exp\left[\frac{-2\pi\zeta}{\sqrt{1-\zeta^2}}\right] = \left(\frac{A}{B}\right)^2$$

Period $$T = \frac{2\pi}{\omega} = \frac{2\pi\tau}{\sqrt{1-\zeta^2}}$$

2.11 Combining Transfer Functions

Most process systems consist of more than one unit operation and have more complex transfer functions, e.g. the distillation column. Transfer functions for the individual elements can be combined to provide a model of the entire system. In this section we look at some examples of combining transfer functions.

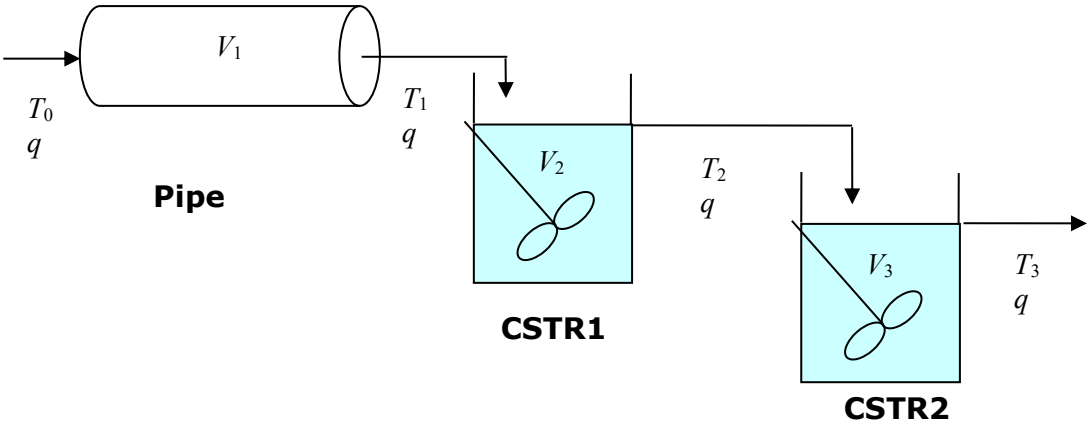
2.11.1 Series



$$\bar{y} = G_1\bar{x} \text{ and } \bar{z} = G_2\bar{y} \rightarrow \bar{z} = G_1G_2\bar{x}$$

Example 2.4

Water flows at a constant rate of 1×10^{-3} m^3/s through a pipe of diameter 0.04 m and length 20 m, before passing into two well-mixed tanks in series, each of constant volume 0.1 m^3. Both tanks and the pipe are well-insulated. Plug flow may be assumed in the pipe section. We will develop a block diagram relating the outlet temperature from the second stirred tank to disturbances in the pipe inlet temperature.



Pipe:

$$u = \frac{4q}{\pi D^2} = \frac{4\times10^{-3}}{\pi\times0.04^2} \approx 0.8 \text{ m/s} \rightarrow t_d = \frac{L}{u} = \frac{20}{0.8} = 25 \text{ s} \rightarrow \bar{T}_1 = \bar{T}_0 e^{-25s}$$

CST 1:

$$\rho V_2 c_p \frac{dT_2}{dt} = \rho q c_p (T_1 - T_2) \rightarrow \tau_2 \frac{dT_2}{dt} + T_2 = T_1 \text{ where } \tau_2 = \frac{V_2}{q} = \frac{0.1}{0.001} = 100 \text{ s}$$

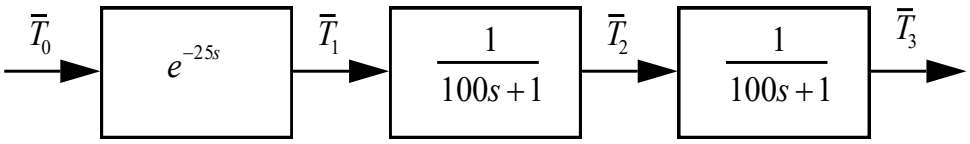
So

$$\frac{\bar{T}_2}{\bar{T}_1} = \frac{1}{\tau_2 s + 1} = \frac{1}{100s + 1}$$

CST 2: Is the same as CST 1, so

$$\frac{\bar{T}_3}{\bar{T}_2} = \frac{1}{100s + 1}$$

The block diagram is



$$\frac{\bar{T}_3}{\bar{T}_0} = \frac{\bar{T}_1}{\bar{T}_0}\frac{\bar{T}_2}{\bar{T}_1}\frac{\bar{T}_3}{\bar{T}_2}$$

Therefore, to obtain the transfer function relating T_3 to T_0 the transfer functions are multiplied:

$$\frac{\bar{T}_3}{\bar{T}_0} = \frac{e^{-25s}}{(100s+1)^2}$$

which is a second order system with a delay.

Q: Does it matter what order the units are in?

Q: How do you invert this to get the response in the time domain? Consider a unit step change in T_0, so

$$\bar{T}_0 = \frac{1}{s} \rightarrow \bar{T}_3 = \frac{e^{-25s}}{s(100s+1)^2}$$

What does the term e^{-25s} do?

Find partial fractions:

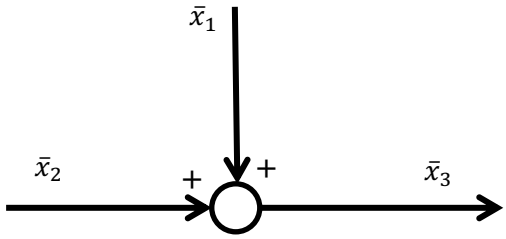
$$\bar{f}(s) = \frac{1}{s(100s+1)^2} = \frac{1}{s} - \frac{100}{100s+1} - \frac{100}{(100s+1)^2} = \frac{1}{s} - \frac{1}{s+1/100} - \frac{1/100}{(s+1/100)^2}$$

$$f(t) = 1 - e^{-t/100} - \frac{t}{100}e^{-t/100}$$

So the response is

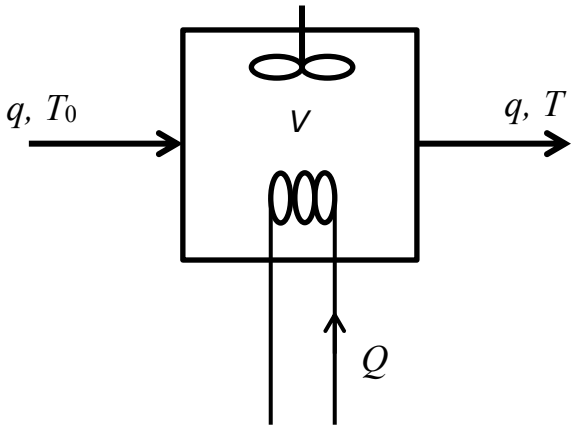
$$\begin{cases} T_3' = 0, \ t < 25 \ \text{s} \\ T_3' = 1 - e^{-(t-25)/100} - \dfrac{(t-25)}{100}e^{-(t-25)/100}, t > 25 \ \text{s} \end{cases}$$

2.11.2 Summation or subtraction



$$\bar{x}_3 = \bar{x}_1 + \bar{x}_2$$

Example 2.5, Heated tank of Section 2.2.5

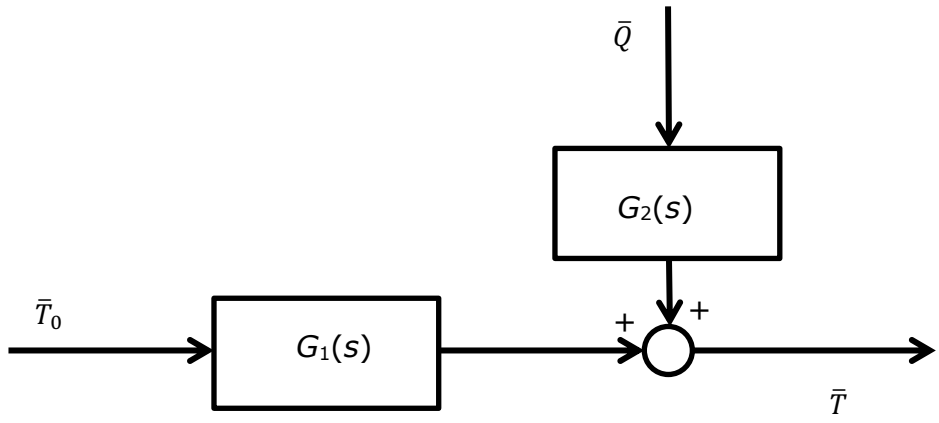


$$\tau\frac{dT}{dt} + T = T_0 + \frac{Q}{\rho q c_p}$$

Take the Laplace transform (deviation variables):

$$(\tau s + 1)\bar{T} = \bar{T}_0 + \frac{\bar{Q}}{\rho q c_p} \rightarrow \bar{T} = \frac{\bar{T}_0}{(\tau s + 1)} + \frac{\bar{Q}}{\rho q c_p (\tau s + 1)}$$

This gives us two transfer functions, one relating changes in T to changes in T_0 with Q unchanged, the other relating changes in T to changes in Q with T_0 unchanged. When both T_0 and Q can change at the same time, the two contributions are added together. The block diagram is as follows.

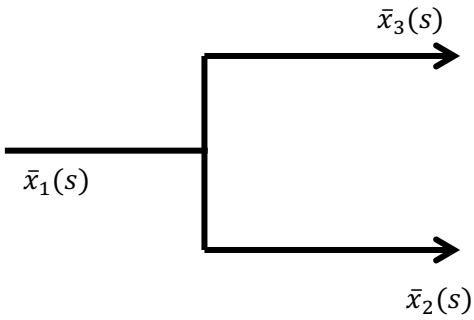


Here

$$G_1(s) = \frac{1}{(\tau s + 1)}, \quad G_2(s) = \frac{1}{\rho q c_p(\tau s + 1)}$$

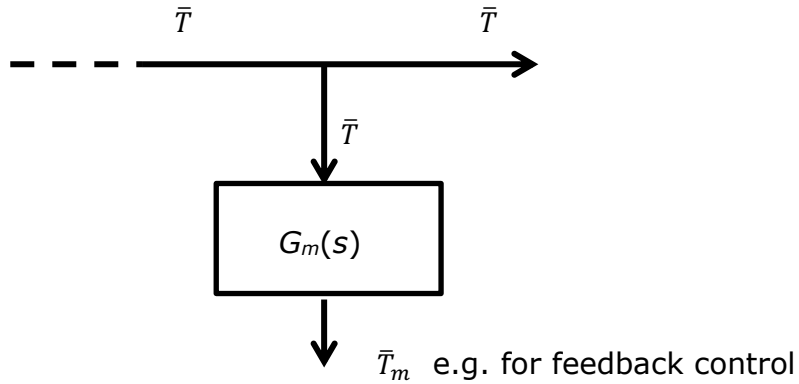
$$\bar{T} = G_1\bar{T}_0 + G_2\bar{Q}$$

2.11.3 Duplicating a signal



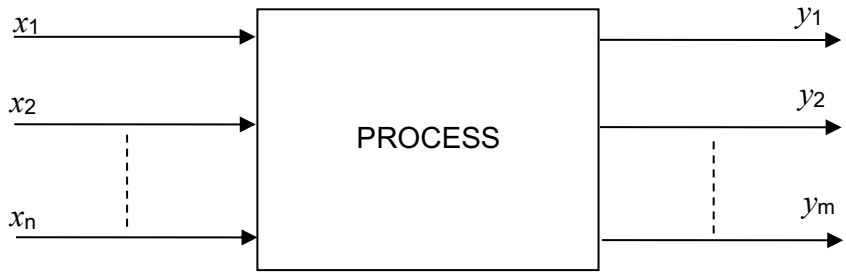
$$\bar{x}_1(s) = \bar{x}_2(s) = \bar{x}_3(s)$$

Example: measuring an outlet temperature to feed back for control.



2.12 Multiple Input Multiple Output (MIMO)

Most real processes consist of multiple inputs that affect multiple outputs e.g. the distillation column in example 1.3. Each input will have an individual influence on each output, and to define the system dynamics, all the transfer functions for the system must be found and combined in a block diagram. This can be done simply by using matrix form.



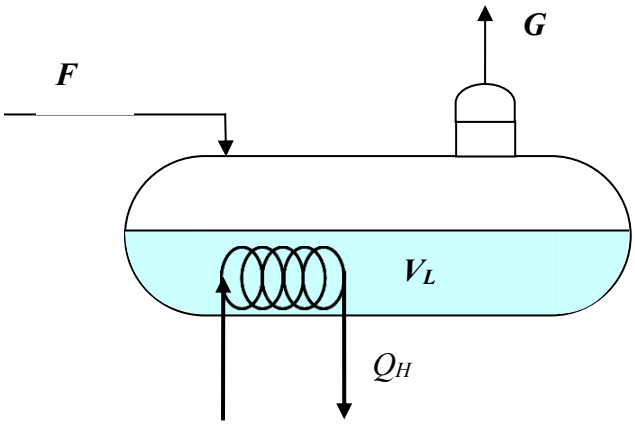
$$\begin{bmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_m \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} & \dots & G_{1n} \\ G_{21} & G_{22} & \dots & G_{2n} \\ \vdots & \vdots & \dots & \vdots \\ G_{m1} & G_{m2} & \dots & G_{mn} \end{bmatrix} \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_n \end{bmatrix}$$

m *m* x *n* *n*

$$\bar{y}_1 = G_{11}\bar{x}_1 + G_{12}\bar{x}_2 + \cdots + G_{1n}\bar{x}_n$$

etc.

Example 2.6 Boiler



Water at 160°C is fed to a boiler operating at 8 bara, saturation temperature 170.4 ºC. *F* and *G* are mass flow rates, Q_H is rate of heating, and V_L is volume of liquid in the boiler. We will develop the transfer function matrix and block diagram that relates the inputs *F* and Q_H to the outputs V_L and *G*.

From steam tables we can find the properties of both phases:

$$\hat{H}_F = 675.5 \text{ kJ/kg}, \hat{H}_L = 720.9 \text{ kJ/kg}, \hat{H}_G = 2768.3 \text{ kJ/kg}, \hat{U}_L = 720.0 \text{ kJ/kg}, \hat{U}_G = 2576.0 \text{ kJ/kg},$$
$$\rho_L = 896.9 \text{ kg/m}^3, \quad \rho_G = 4.2 \text{ kg/m}^3$$

Assume that the total volume is constant: $\frac{d}{dt}(V_L + V_G) = 0$

The mass balance is:

$$\frac{d}{dt}(\rho_L V_L + \rho_G V_G) = (\rho_L - \rho_G)\frac{dV_L}{dt} = F - G$$

The energy balance is:

$$\frac{dU}{dt} = \frac{d}{dt}\left(\rho_L \hat{U}_L V_L + \rho_G \hat{U}_G V_G\right) = \left(\rho_L \hat{U}_L - \rho_G \hat{U}_G\right)\frac{dV_L}{dt} = F\hat{H}_F - G\hat{H}_G + Q_H$$

Solving for $\frac{dV_L}{dt}$ and G:

$$\frac{dV_L}{dt} = \frac{F(\hat{H}_G - \hat{H}_F) - Q_H}{\rho_L(\hat{H}_G - \hat{U}_L) - \rho_G(\hat{H}_G - \hat{U}_G)}$$

$$G = \frac{Q_H(\rho_L - \rho_G) + F\left(\rho_L\left(\hat{H}_F - \hat{U}_L\right) + \rho_G\left(\hat{U}_G - \hat{H}_F\right)\right)}{\rho_L(\hat{H}_G - \hat{U}_L) - \rho_G(\hat{H}_G - \hat{U}_G)}$$

Substituting values:

$$\frac{dV_L}{dt} = \frac{F(2768.3 - 675.5) - Q_H}{896.9(2768.3 - 720.0) - 4.2(2768.3 - 2576.0)} = 1.14 \times 10^{-3}F - 5.45 \times 10^{-7}Q_H$$

Take the Laplace transform (remember – deviation variables):

$$s\bar{V}_L = 1.14 \times 10^{-3}\bar{F} - 5.45 \times 10^{-7}\bar{Q}_H$$

$$G = \frac{Q_H(896.9 - 4.2) + F\left(896.9(675.5 - 720) + 4.2(2576.0 - 675.5)\right)}{896.9(2768.3 - 720) - 4.2(2768.3 - 2576.0)} = -0.0174F + 4.86 \times 10^{-4}Q_H$$

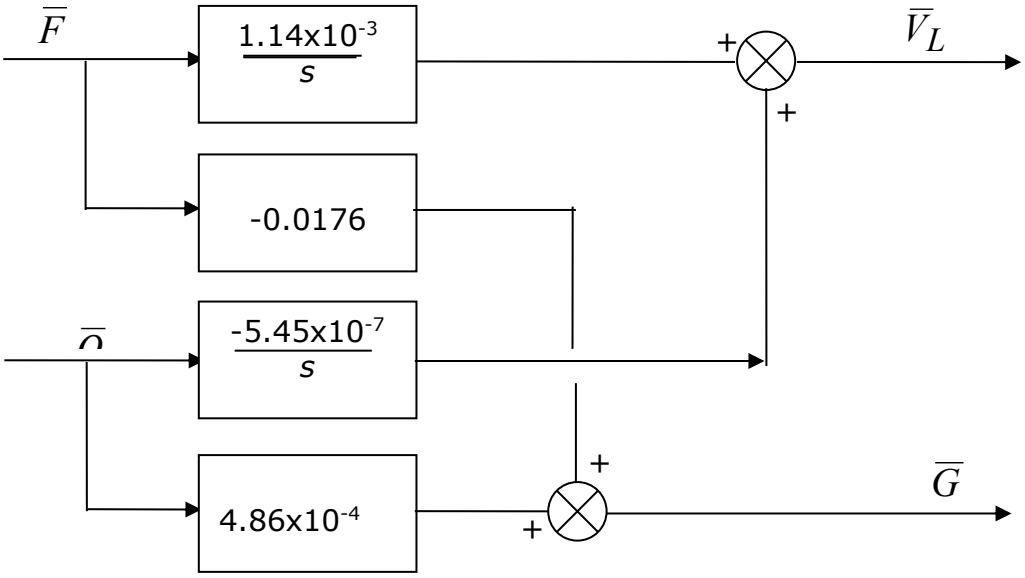
Take the Laplace transform (remember – deviation variables):

$$\bar{G} = -0.0174\bar{F} + 4.86 \times 10^{-4}\bar{Q}_H$$

In matrix form:

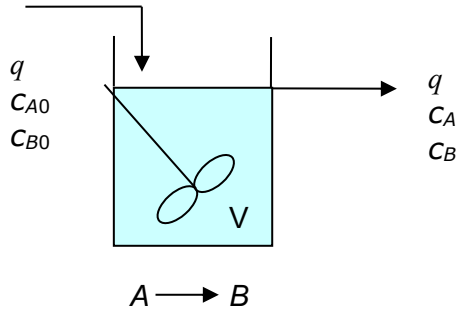
$$\begin{bmatrix}\bar{V}_L \\ \bar{G}\end{bmatrix} = \begin{bmatrix}\frac{1.14 \times 10^{-3}}{s} & \frac{-5.45 \times 10^{-7}}{s} \\ -0.0174 & 4.86 \times 10^{-4}\end{bmatrix}\begin{bmatrix}\bar{F} \\ \bar{Q}_H\end{bmatrix}$$

The block diagram is:



Example 2.7

A CSTR of constant volume V is used for the first order irreversible reaction of component A to component B. The feed to the reactor, q m^3/h, is a mixture of A and B. Develop the transfer function matrix and block diagram that relates the input concentrations c_{A0} and c_{B0} to the outlet concentrations c_A and c_B.



Rate of reaction for A: $r = -kc_A$

Residence time $\tau = V/q$

Write down mass balances, and take Laplace transforms (remember – deviation variables):

on component *A*:

$$V\frac{dc_A}{dt} = qc_{A0} - qc_A - Vkc_A \;\rightarrow\; \tau\frac{dc_A}{dt} + (1+\tau k)c_A = c_{A0}$$

on component *B*:

$$V\frac{dc_B}{dt} = qc_{B0} - qc_B + Vkc_A \;\rightarrow\; \tau\frac{dc_B}{dt} + c_B - \tau kc_A = c_{B0}$$

Then

$$(\tau s + 1 + \tau k)\bar{c}_A = \bar{c}_{A0} \;\rightarrow\; \bar{c}_A = \frac{\bar{c}_{A0}}{(\tau s + 1 + \tau k)}$$

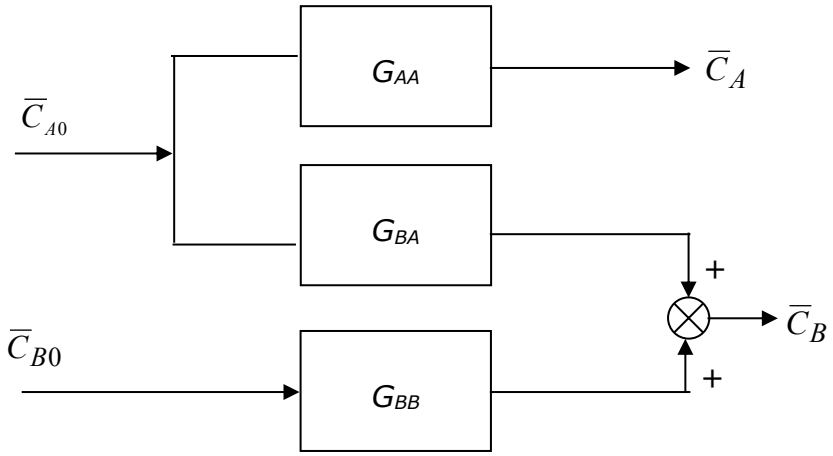
$$\rightarrow (\tau s+1)\bar{c}_B - \tau k\bar{c}_A = \bar{c}_{B0} \rightarrow \bar{c}_B = \frac{\bar{c}_{B0}}{(\tau s+1)} + \frac{\tau k\bar{c}_{A0}}{(\tau s+1)(\tau s+1+\tau k)}$$

In matrix form:

$$\begin{bmatrix}\bar{c}_A\\ \bar{c}_B\end{bmatrix} = \begin{bmatrix}G_{AA} & G_{AB}\\ G_{BA} & G_{BB}\end{bmatrix}\begin{bmatrix}\bar{c}_{A0}\\ \bar{c}_{B0}\end{bmatrix}$$

$$\begin{bmatrix}\bar{c}_A\\ \bar{c}_B\end{bmatrix} = \begin{bmatrix}\left(\frac{1}{\tau s+1+\tau k}\right) & 0\\ \left(\frac{\tau k}{(\tau s+1)(\tau s+1+\tau k)}\right) & \left(\frac{1}{\tau s+1}\right)\end{bmatrix}\begin{bmatrix}\bar{c}_{A0}\\ \bar{c}_{B0}\end{bmatrix}$$

The block diagram is shown below:



2.13 System Stability

We have seen that the stability of a second order system is controlled by the nature of the poles of the transfer function. The same is true of the first order lag:

$$G(s) = \frac{K_p}{\tau s+1}$$

$$\bar{x} = \frac{1}{s} \rightarrow \bar{y} = \frac{K_p}{s(\tau s+1)} = K_p\left[\frac{1}{s} - \frac{\tau}{\tau s+1}\right] \rightarrow y' = K_p\left[1-e^{-t/\tau}\right]$$

The 1/*s* term comes from the choice of input, and leads to a constant term in the output. The characteristic equation is

$$\text{Denominator of transfer function } = D(s) = \tau s+1 = 0 \rightarrow p = -\frac{1}{\tau}$$

implying stable behaviour.

2.13.1 Poles

The transfer function can often be expressed as the ratio of two polynomials in the Laplace variable *s*:

$$G(s) = \frac{N(s)}{D(s)}$$

where, $N(s)$ and $D(s)$ are polynomials of the orders m and n respectively. For all real physical systems $m < n$ i.e. the denominator will be of a higher order than the numerator.

2.13.2 General Form

An example transfer function which displays all types of poles is:

$$G(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s-p_1)(s-p_2)^n(s-p_3)(s-p_3^*)}$$

p_1 = *unique or distinct root*
p_2 = *repeated root*
p_3 = *complex root*
p_3^* = *complex conjugate*

G is real for real *s*, so complex poles come in complex conjugate pairs.

Not all types of roots will be present in all problems.

To find the roots the characteristic equation, $D(s) = 0$, must be solved.

Taking partial fractions:

$$G(s) = \frac{C_1}{s-p_1} + \left[\frac{C_{21}}{s-p_2} + \frac{C_{22}}{(s-p_2)^2} + \frac{C_{23}}{(s-p_2)^3} + \cdots + \frac{C_{2n}}{(s-p_2)^n}\right] + \frac{C_3}{s-p_3} + \frac{C_3^*}{s-p_3^*}$$

The sum of the last two terms is real.

2.13.3 Response to a Unit Impulse

$$x'(t) = \delta(t) \rightarrow \bar{x} = 1 \rightarrow \bar{y} = G(s)$$

$$y' = C_1 e^{p_1 t} + \left[C_{21}e^{p_2 t} + C_{22}te^{p_2 t} + C_{23}\frac{t^2}{2!}e^{p_2 t} + \cdots + C_{2n}\frac{t^{n-1}}{(n-1)!}e^{p_2 t}\right] + C_3 e^{p_3 t} + C_3^* e^{p_3^* t}$$

For a real root, distinct or repeated:

$p < 0 \rightarrow$ decaying exponential, stability

$p > 0 \rightarrow$ increasing exponential, instability

For a pair of complex roots, write, with *a*, *b*, α and β real,

$$p_3 = a + ib,\ p_3^* = a - ib,\ C_3 = \alpha + i\beta,\ C_3^* = \alpha - i\beta$$

Then

$$C_3 e^{p_3 t} + C_3^* e^{p_3^* t} = (\alpha + i\beta)e^{at}\big(\cos(bt) + isin(bt)\big) + (\alpha - i\beta)e^{at}\big(\cos(bt) - isin(bt)\big)$$

$$= 2e^{at}[\alpha\cos(bt) - \beta\sin(bt)] = Ce^{at}\cos(bt + \phi)$$

$$\text{where } C = 2\sqrt{\alpha^2 + \beta^2},\ \tan(\phi) = \frac{\beta}{\alpha}$$

Hence the response is oscillatory, and

$a < 0$, i.e. $\text{Re}(p_3) < 0 \rightarrow$ decaying exponential, stability

$a > 0$, i.e. $\text{Re}(p_3) > 0 \rightarrow$ increasing exponential, instability

2.13.4 Zeros

A zero is the root of the numerator of the transfer function. Zeros do not influence the exponents and therefore the system stability. They do alter the constants in the equation which gives the transient solution, and so they influence the dynamic response.

2.13.5 Example 2.7

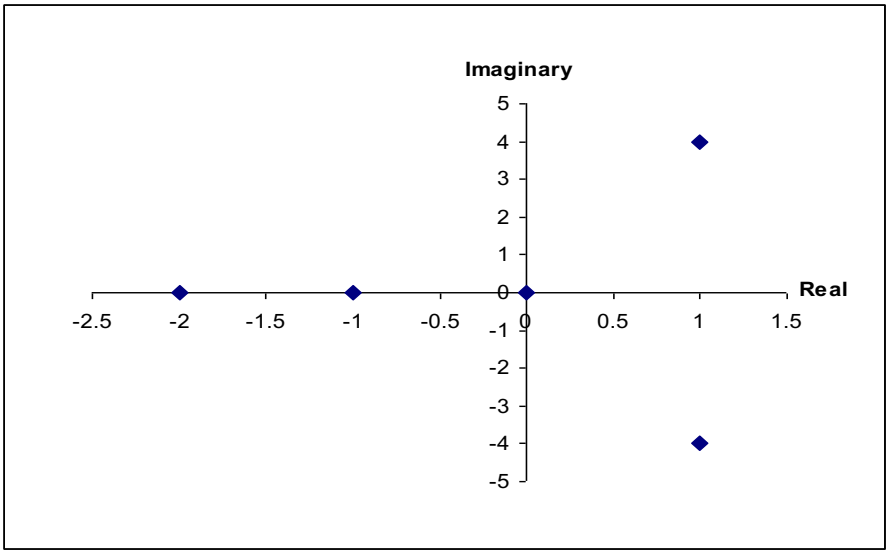
Consider a process with the following transfer function

$$G(s) = \frac{N(s)}{D(s)} = \frac{1}{34s + 81s^2 + 60s^3 + 14s^4 + 2s^5 + s^6}$$

Find the poles and plot them in the complex plane. Determine the unit impulse response of this system.

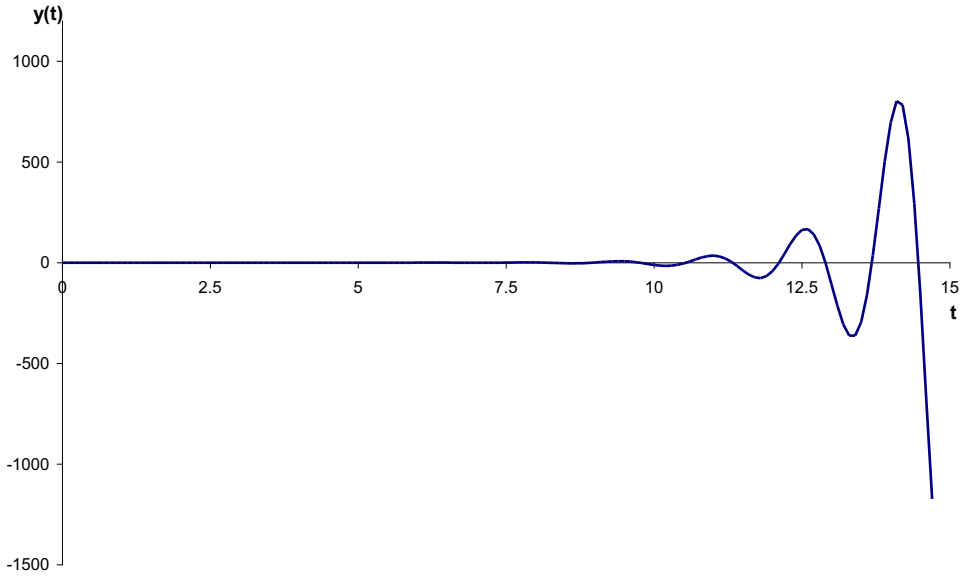
Solution

Factorising $D(s) = s(s+1)^2(s+2)(s-1-4i)(s-1+4i)$



Inverting the transfer function, for $\bar{x} = 1$

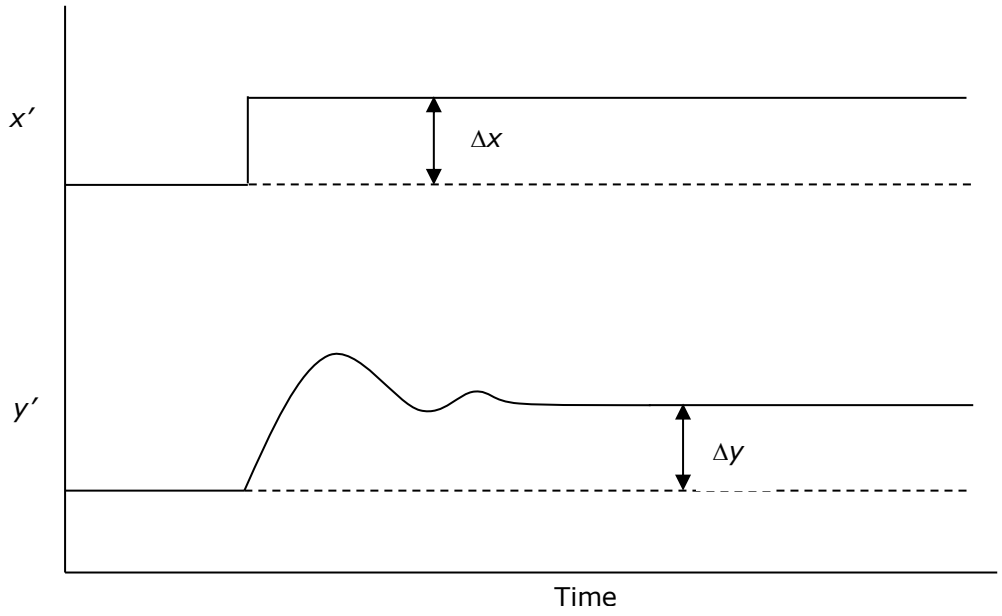
$$L^{-1}\left[\frac{N(s)}{D(s)}\right] = \frac{1}{34} - \frac{e^{-2t}}{50} - \frac{e^{-t}}{100} - \frac{te^{-t}}{20} + \frac{e^{t}(\cos(4t) - 0.25\sin(4t))}{1700}$$



.

2.14 Steady State Process Gain

For a stable system, a step change in $x(t)$ will result eventually in a new steady state. This can be found without evaluating the transient response using the Final Value Theorem.



Steady state gain = $K_P = \frac{\Delta Y}{\Delta X}$

It's possible to evaluate the steady state process gain by setting $s = 0$ in the transfer function.

Example 2.9

A process has a transfer function: $G(s) = \dfrac{10e^{-5s}}{(s+1)(s+5)(s+10)}$

This represents a dead time and three first order lags in series. The poles of *G* are negative, therefore the system is stable. Find the new steady state when

$$x'(t) = 1 - e^{-t/10} \rightarrow \bar{x} = \frac{1}{s} - \frac{1}{s+0.1} = \frac{1}{s(10s+1)}$$

$$\Delta y = \lim_{s\to 0} sG(s)\bar{x} = \lim_{s\to 0} \frac{10e^{-5s}}{(10s+1)(s+1)(s+5)(s+10)} = \frac{10}{1\times 1\times 5\times 10} = 0.2$$

Equivalently, evaluate *G*(0) = 0.2, and multiply by the steady value of *x'* = 1.

2.15 Linearisation

Many real processes although nonlinear, can be successfully modelled using linear approximations, as the range of operation of interest is normally close to a steady state.

An operator F is linear if the following relationship holds (a and b are constants):

$$F(ax_1 + bx_2) = aF(x_1) + bF(x_2)$$

e.g. if $F(x) = kx$,

$$F(ax_1 + bx_2) = k(ax_1 + bx_2) = kax_1 + kbx_2 = aF(x_1) + bF(x_2)$$

By this criterion $F(x) = kx^{1/2}$ is nonlinear:

$$k(ax_1 + bx_2)^{1/2} \neq k(ax_1)^{1/2} + k(bx_2)^{1/2}$$

A nonlinear expression can be linearised by a Taylor expansion:

$$f(x) = f(x_s + x') = f(x_s) + x'f'(x_s) + O((x')^2)$$

In some cases it's easier to expand brackets, or use the Binomial expansion, e.g.

$$\left(x_s + x'\right)^2 = x_s^2 + 2x_s x' + (x')^2 \approx x_s^2 + 2x_s x'$$

$$\left(x_s + x'\right)^\lambda = x_s^\lambda \left(1 + \frac{x'}{x_s}\right)^\lambda \approx x_s^\lambda \left(1 + \frac{\lambda x'}{x_s}\right) = x_s^\lambda + \lambda x_s^{\lambda-1} x'$$

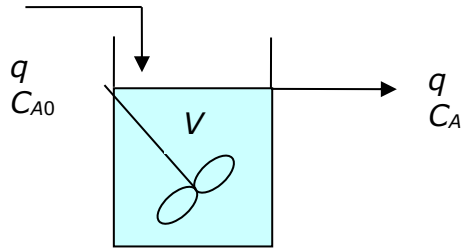
If the process dynamics is written as

$$F(y) = G(x)$$

we want *F* and *G* to be linear operators.

Example 2.10

A CSTR with constant volume V and a flowrate of q converts *A* to *B* in an irreversible reaction. In one case the reaction is first order with respect to *A* and in the second it is second order with respect to *A*. Are the systems linear in C_A?



CSTR with first order reaction

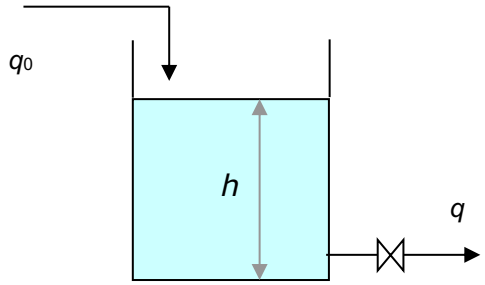
$$V\frac{dC_A}{dt} = q(C_{Ao} - C_A) - VkC_A$$

CSTR with second order reaction

$$V\frac{dC_A}{dt} = q(C_{Ao} - C_A) - VkC_A^2$$

Example 2.11

Water flows into a tank at a volumetric flow rate q_0. The outlet flow rate q is determined by the depth of liquid in the tank, h, and Bernoulli's equation gives $q = k\sqrt{h}$. The cross sectional area of the tank is A.



At steady state:

$$q_{0s} = q_s = k\sqrt{h_s}$$

The volume balance is

$$A\frac{dh}{dt} = q_0 - q = q_0 - k\sqrt{h}$$

which is nonlinear. Write

$$q_0 = q_{0s} + q_0',\ q = q_s + q',\ h = h_s + h'$$

and use the binomial theorem

$$(h_s + h')^{1/2} = (h_s)^{1/2}\left(1 + \frac{h'}{h_s}\right)^{1/2} = (h_s)^{1/2}\left(1 + \frac{h'}{2h_s} + \cdots\right) \approx \sqrt{h_s} + \frac{h'}{2\sqrt{h_s}}$$

Then the ODE becomes, approximately,

$$A\frac{dh'}{dt} = q_{0s} + q_0' - k\sqrt{h_s} - \frac{kh'}{2\sqrt{h_s}} = q_0' - \frac{kh'}{2\sqrt{h_s}}$$

The steady state terms have cancelled out, as they must. This is now a linear equation; indeed it is a first order lag:

$$\frac{2A\sqrt{h_s}}{k}\frac{dh'}{dt} + h' = \frac{2\sqrt{h_s}}{k}q_0'$$

If $q_0' = \Delta q_0$,a constant, the solution is

$$h' = \frac{2\sqrt{h_s}}{k}\Delta q_0\left[1 - e^{-kt/(2A\sqrt{h_s})}\right]$$

This will be accurate enough when *h*' is small enough, but the asymptotic value h_∞' given by the linearization

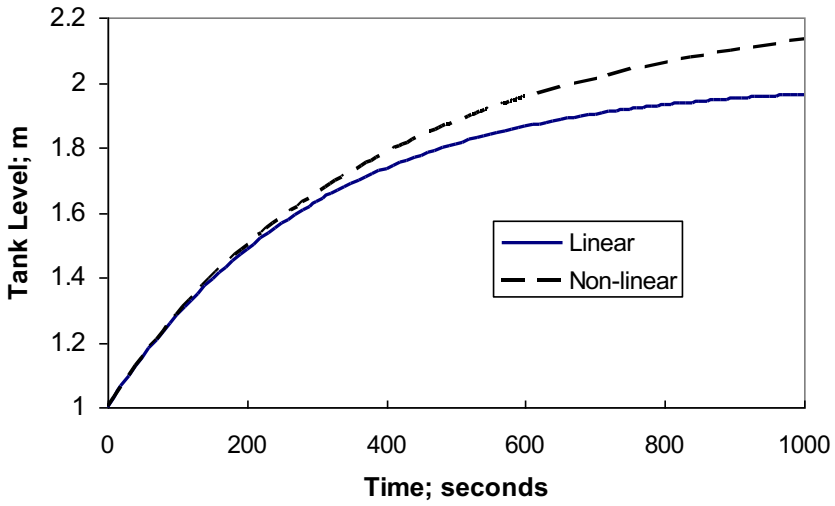
$$h_\infty' = \frac{2\sqrt{h_s}}{k}\Delta q_0$$

differs from the exact answer given by

$$q_{0s} + \Delta q_0 = k\sqrt{h_s + h_\infty'} \rightarrow h_s + h_\infty' = \left(\frac{q_{0s} + \Delta q_0}{k}\right)^2 = \left(\frac{q_{0s}}{k}\right)^2 + 2\left(\frac{q_{0s}}{k}\right)\left(\frac{\Delta q_0}{k}\right) + \left(\frac{\Delta q_0}{k}\right)^2$$

$$h_\infty' = \frac{2\sqrt{h_s}}{k}\Delta q_0 + \left(\frac{\Delta q_0}{k}\right)^2$$

Take A = 3 m^2, the initial steady state to be q_0 = 0.02 m^3/s and k = 0.02 m$^{5/2}$/s so that h_s = 1 m. Take Δq_0 = 0.01 m^3/s. The exact response and the linear approximation are plotted below.



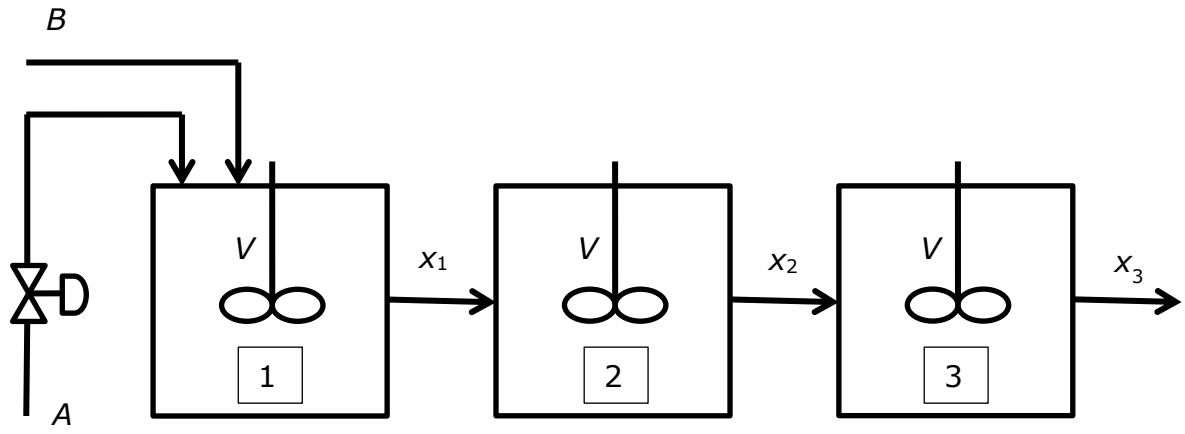
Q. What is the steady state gain for the linear response?

Exercise. Transform the nonlinear ODE by the substitution $h = z^2$, linearise the ODE, and compare the solution to the one above.

Exercise. Find the exact solution.

Example 2.12. Three tank mixing process

This example is taken from Marlin. It will be used to demonstrate controller behaviour in the next section.



Three tanks of equal volume, 35 m^3, are used to mix two streams, *A* and *B*. Stream *A* is 100% component *Z*, an inert tracer, which has the same density as water. Stream *B* has volume percent x_B of *Z* in water. Initially (design conditions) the flow rate of stream *B*, q_{Bs} = 6.86 m^3/min, x_{Bs} = 2%, the flow rate of stream A, q_{As} = 0.14 m^3/min, with the linear valve being 50% open, and the volume fraction of *Z* in each of the three tanks is $x_{1s} = x_{2s} = x_{3s}$ = 4%.

At design conditions, the mean residence time in each tank is τ = 5 minutes.

The valve has a linear characteristic, i.e. the flow rate is proportional to the stem position *v* (% open):

$$q_A = kv$$

where $k = 2.8\times10^{-3}$ m^3/min %, so that $q_A = 0.14$ m^3/min when $v = 50\%$ open.

The volume balance on Z in tank 1 is

$$V\frac{dx_1}{dt} = 100q_A + q_B x_B - (q_A + q_B)x_1$$

We will suppose that the main disturbance is in x_B, and that $q_B = q_{Bs}$. Introduce deviation variables:

$$q_A = q_{As} + q_A',\ \ q_B = q_{Bs},\ \ x_B = x_{Bs} + x_B',\ \ x_1 = x_{1s} + x_1'$$

The ODE becomes

$$V\frac{dx_1'}{dt} = 100(q_{As} + q_A') + q_{Bs}(x_{Bs} + x_B') - (q_{As} + q_A' + q_{Bs})(x_{1s} + x_1')$$

This gives

$$V\frac{dx_1'}{dt} = [100q_{As} + q_{Bs}x_{Bs} - (q_{As} + q_{Bs})x_{1s}] - (q_{As} + q_{Bs})x_1' + q_{Bs}x_B' + (100 - x_{1s})q_A' - q_A'x_1'$$

The term in square brackets is the steady state volume balance, so vanishes, and the last term is a product of deviation variables, so is ignored. This leaves, approximately,

$$V\frac{dx_1'}{dt} = -(q_{As} + q_{Bs})x_1' + q_{Bs}x_B' + (100 - x_{1s})q_A'$$

or

$$\tau\frac{dx_1'}{dt} + x_1' = \frac{q_{Bs}}{q_{As} + q_{Bs}}x_B' + \frac{100 - x_{1s}}{q_{As} + q_{Bs}}q_A' \quad \text{where} \quad \tau = \frac{V}{q_{As} + q_{Bs}}$$

The volume balance on Z in tank 2 is

$$V\frac{dx_2}{dt} = (q_A + q_B)x_1 - (q_A + q_B)x_2 = (q_A + q_B)(x_1 - x_2) = (q_A + q_B)(x_1' - x_2')$$
$$\approx (q_{As} + q_{Bs})(x_1' - x_2')$$

so that

$$\tau\frac{dx_2'}{dt} + x_2' = x_1'$$

Similarly

$$\tau\frac{dx_3'}{dt} + x_3' = x_2'$$

For the valve:

$$q_A' = kv'$$

Putting in the numbers and taking Laplace transforms gives the linearised transfer function model:

$$\bar{x}_3 = \frac{0.98}{(5s+1)^3}\bar{x}_B + \frac{0.038}{(5s+1)^3}\bar{v}$$

Let's calculate the response to a step in valve position Δv.

$$\bar{x}_3 = \frac{\Delta v}{s}\frac{0.038}{(5s+1)^3} = 0.038\Delta v\left[\frac{1}{s} - \frac{5}{5s+1} - \frac{5}{(5s+1)^2} - \frac{5}{(5s+1)^3}\right]$$
$$= 0.038\Delta v\left[\frac{1}{s} - \frac{1}{s+1/5} - \frac{1}{5}\frac{1}{(s+1/5)^2} - \frac{1}{25}\frac{1}{(s+1/5)^3}\right]$$

so that

$$x_3' = 0.038\Delta v\left[1 - \left(1 + \frac{t}{5} + \frac{t^2}{50}\right)e^{-t/5}\right]$$

We will use this in the next section.